

Machine Learning 2024-2045
Homework 2 - Group 127
(ist1106900, ist1106937)
I. - Pen-and-Paper

Pen and Paper

①

	x_i	x_L	z_i	z_L
n_1	A	0	1 (P)	N
n_2	B	1	1 (P)	N
n_3	A	1	1 (P)	P
n_4	A	0	1 (P)	N
n_5	B	0	0 (N)	P
n_6	B	0	0 (N)	P
n_7	A	1	0 (N)	P
n_8	B	1	0 (N)	N

$$d(n_i, n_j) = \text{Hamming}(n_i, n_j) = (1+1) = 2$$

$d(n_i, n_j)$	n_1	n_2	n_3	n_4	n_5	n_6	n_7	n_8
n_1	2	2	1	0	1	1	1	2
n_2	2	2	1	1	1	1	0	0
n_3	1	1	2	1	2	2	0	1
n_4	0	1	1	2	1	1	1	2
n_5	1	1	2	1	2	0	2	1
n_6	1	1	2	1	0	2	2	1
n_7	1	1	0	1	2	2	2	1
n_8	2	0	1	2	1	1	1	2

	True label	Predicted Label
$n_1 \rightarrow$	P	N (FN)
$n_2 \rightarrow$	P	N (FN)
$n_3 \rightarrow$	P	P (TP)
$n_4 \rightarrow$	P	N (FN)
$n_5 \rightarrow$	N	P (FP)
$n_6 \rightarrow$	N	P (FP)
$n_7 \rightarrow$	N	P (FP)
$n_8 \rightarrow$	N	N (TN)

Weighted mode = $d_1 + d_2 + d_3 + \dots$

$$\begin{aligned} n_1 &+ 1 + 0 = 1 & n_1 &- 1 + 1 + 1 = 3 \\ n_2 &+ 1 = 1 & n_2 &- 1 + 1 + 1 = 3 \\ n_3 &+ 1 + 1 + 1 = 3 & n_3 &- 0 + 1 = 1 \\ n_4 &+ 0 + 1 = 1 & n_4 &- 1 + 1 + 1 = 3 \\ n_5 &+ 1 + 1 + 1 = 3 & n_5 &- 0 + 1 = 1 \\ n_6 &+ 1 + 1 + 1 = 3 & n_6 &- 0 + 1 = 1 \\ n_7 &+ 1 + 1 + 0 + 1 = 3 & n_7 &- 1 = 1 \\ n_8 &+ 0 + 1 = 1 & n_8 &- 1 + 1 + 1 = 3 \end{aligned}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{1}{1+3} = \frac{1}{4} = \frac{1}{4} \rightarrow R$$

$$\text{Precision} = \frac{1}{1+3} = \frac{1}{4} \rightarrow P$$

$$\begin{aligned} F1\text{-measure} &= 2 \times \frac{P \times R}{P + R} = 2 \times \left(\frac{\frac{1}{4} \times \frac{1}{4}}{\frac{1}{4} + \frac{1}{4}} \right) \\ &= \frac{1}{4} = 0.25 \end{aligned}$$

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② To triple the F1-measure, we can change the distance metric to 0 (A or B), instead. Since A is more associated with positive values and B to negative values, it would make sense to give more weight to them.

So, we are going to use a new weighted Hamming Distance and $k=3$

$$D(n_i, n_j) = w_1 d_1(n_i, n_j) + w_2 d_2(n_i, n_j), \quad w_1=1 \quad w_2=0$$

$d(n_i, n_j)$	n_1	n_2	n_3	n_4	n_5	n_6	n_7	n_8	$\hat{z}_i$	z_i
n_1	/	1	0	0	1	1	0	1	P	P (TP)
n_2	1	/	1	1	0	0	1	0	N	P (FP)
n_3	0	1	/	0	1	1	0	1	P	P (TP)
n_4	0	1	0	/	1	1	0	1	P	P (TP)
n_5	1	0	1	1	/	0	1	0	N	N (TN)
n_6	1	0	1	1	0	/	1	0	N	N (TN)
n_7	0	1	0	0	1	1	/	1	P	N (FN)
n_8	1	0	1	1	0	0	1	/	N	N (TN)

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{3}{1+3} = \frac{3}{4} \quad \cdot \text{Recall} = \frac{TP}{TP+FN} = \frac{3}{4+3} = \frac{3}{7} = 0.428$$

$$F1\text{-measure} = 2 \cdot \frac{P \times R}{P+R} = \frac{\frac{3}{4} \cdot \frac{3}{7}}{\frac{3}{4} + \frac{3}{7}} \cdot 2 = \frac{3}{4} = 0.75 = 3 \cdot 0.25,$$

conseguimos uma f1-measure 3 vezes maior ✓

③

	y_1	y_2	y_3	z
x_1	A	0	1.1	1
x_2	B	1	0.8	1
x_3	A	1	0.5	1
x_4	A	0	0.9	1
x_5	B	0	1	0
x_6	B	0	0.9	0
x_7	A	1	1.2	0
x_8	B	1	0.9	0
x_9	B	0	0.8	1

$$\{y_1, y_2\} \perp \{y_3\} = P(y_1, y_2, y_3 | z) =$$

$$= P(y_1, y_2 | z) \times P(y_3 | z)$$

$$\text{Bayesian Theorem: } P(z | y) = \frac{P(y | z) \cdot P(z)}{P(y)}$$

$$P(z): \text{ priors: } P(z=1) = 5/9, P(z=0) = 4/9$$

$$P(y_1, y_2 | z): \text{ PMFs}$$

$$P(y_1=A, y_2=0 | z=1) = 2/5 = 0.4$$

$$P(y_1=A, y_2=0 | z=0) = 0$$

$$P(y_1=A, y_2=1 | z=1) = 1/5 = 0.2$$

$$P(y_1=A, y_2=1 | z=0) = 1/4 = 0.25$$

$$P(y_1=B, y_2=0 | z=1) = 1/5 = 0.2$$

$$P(y_1=B, y_2=0 | z=0) = 2/4 = 0.5$$

$$P(y_1=B, y_2=1 | z=1) = 1/5 = 0.2$$

$$P(y_1=B, y_2=1 | z=0) = 1/4 = 0.25$$

$$P(y_3 | z): \text{ PDFs}$$

$$P(A, 0) = \frac{2}{9} \quad P(A, 1) = \frac{2}{9} \quad P(B, 0) = \frac{3}{9} \quad P(B, 1) = \frac{2}{9}$$

$\mu_1 / \sigma_1 \rightarrow$ mean / standard deviation of observations where $z=1$

$\mu_0 / \sigma_0 \rightarrow$ " " " " " " $z=0$

$$\mu_1 = \frac{1.1 + 0.8 + 0.5 + 0.9 + 0.8}{5} = 0.82$$

$$\mu_0 = \frac{1 + 0.9 + 1.2 + 0.9}{4} = 1$$

$$\sigma_0 = \dots = 0.1225$$

$$\sigma_1 = \sqrt{\frac{(1.1-0.82)^2 + (0.8-0.82)^2 + (0.5-0.82)^2 + (0.9-0.82)^2 + (0.8-0.82)^2}{5}} = 0.1939$$

$$N(\mu_3=0.9, \sigma_3=0.03556)$$

$$N(\mu_3=0.82, \sigma_3=0.01939 | P)$$

$$N(\mu_3=1, \sigma_3=0.1225 | N)$$

Pen-and-Paper

(4) Under map assumption, $P(z|x) = \frac{P(z|x) P(z)}{P(x)}$

$$P(P) = 5/9$$

$$P(N) = 4/9$$

For $(A, 1, 0.8)$,

Positive:

$$P(y_1=A|P) = 0.6$$

$$P(y_2=1|P) = 0.4$$

$$P(y_3=0.8|P) = \frac{1}{\sqrt{2\pi} \cdot 0.1939} \cdot e^{\left(-\frac{1}{2 \cdot 0.1939^2} \cdot (0.8-0.82)^2\right)} = 2.046$$

$$P(P|x) \propto P(x|P) \cdot P(P) = P(y_1=A|P) \cdot P(y_2=1|P) \cdot P(y_3=0.8|P) \cdot P(P) = 0.6 \cdot 0.4 \cdot 2.046 \cdot \frac{5}{9} \approx 0.273$$

Negative

$$P(y_1=A|N) = 0.25$$

$$P(y_2=1|N) = 0.7$$

$$P(y_3=0.8|N) = \frac{1}{\sqrt{2\pi} \cdot 0.1225} \cdot e^{\left(-\frac{1}{2 \cdot 0.1225^2} \cdot (0.8-1)^2\right)} \approx 0.8589$$

$$P(N|x) \propto P(x|N) \cdot P(N) = 0.25 \cdot 0.7 \cdot 0.8589 \cdot \frac{4}{9} \approx 0.047$$

Como $P(P|x) > P(N|x)$, $(A, 1, 0.8)$ é classificada como positiva

Para $(B, 1, 1)$:

Positiva:

$$P(y_1=B|P) = 0.4$$

$$P(y_2=1|P) = 0.4$$

$$P(y_3=1|P) = \frac{1}{\sqrt{2\pi} \cdot 0.1939} \cdot e^{\left(-\frac{1}{2 \cdot 0.1939^2} \cdot (1-0.82)^2\right)} = 1.3321$$

$$P(P|x) = 0.4 \cdot 0.4 \cdot \frac{5}{9} \cdot 1.3321 = 0.1189$$

Negativa:

$$P(y_1=B|N) = 0.75$$

$$P(y_2=1|N) = 0.7$$

$$P(y_3=1|N) = \frac{1}{\sqrt{2\pi} \cdot 0.1225} \cdot e^{\left(-\frac{1}{2 \cdot 0.1225^2} \cdot (1-1)^2\right)} = 3.2567$$

$$P(N|x) = 0.75 \cdot 0.7 \cdot \frac{4}{9} \cdot 3.2567 \approx 0.9428$$

Como $P(N|x) > P(P|x)$, $(B, 1, 1)$ é classificada como negativa.

Para $(0, 0, 0.9)$

Positiva:

$$P(y_1 = 0 | P) = 0.4$$

$$P(y_2 = 0 | P) = 0.6$$

$$P(y_3 = 0.9 | P) = \frac{1}{\sqrt{2\pi} \cdot 0.1934} \cdot e^{\left(-\frac{1}{2 \cdot 0.1934^2} \cdot (0.9 - 0.82)^2\right)} = 1.8845$$

$$P(P|x) = 0.4 \times 0.6 \times \frac{5}{7} \times 1.8845$$

$$\approx 0.2514$$

Negativa:

$$P(y_1 = 0 | N) = 0.15$$

$$P(y_2 = 0 | N) = 0.9$$

$$P(y_3 = 0.9 | N) = \frac{1}{\sqrt{2\pi} \cdot 0.1225} \cdot e^{\left(-\frac{1}{2 \cdot 0.1225^2} \cdot (0.9 - 1)^2\right)} = 2.334$$

$$P(N|x) = 0.5 \times 0.9 \times \frac{2}{7} \times 2.334$$

$$\approx 0.3884$$

Como $P(N|x) > P(P|x)$, $(0, 0, 0.9)$ é classificada como negativa

⑤ Positive (P) sentences: "Amazing run" "I like it"

Negative (N) sentences: "Too Tired" "Bad run"

Existem 8 palavras $V=8$

$$\text{Class P: } p(t_i | P) = \frac{\text{freq}(t_i \text{ in } P) + 1}{5 + 8 = 13}$$

P class: $N_P = 5$

N class: $N_N = 4$

$$\text{Class N: } p(t_i | N) = \frac{\text{freq}(t_i \text{ in } N) + 1}{4 + 8 = 12}$$

Positive:

$$P("I" | P) = \frac{1+1}{5+8} = \frac{2}{13} \approx 0.1538$$

$$P("to" | P) = 0.0769$$

$$P("like" | P) = 0.1538$$

$$P("run" | P) = 0.1538$$

Negative:

$$P("I" | N) = 0.0833$$

$$P("like" | N) = 0.0833$$

$$P("to" | N) = 0.0833$$

$$P("run" | N) = 0.1667$$

$$P("I like to run" | P) = 0.1538 \cdot 0.0769 \cdot 0.1538 = 0.0028$$

$$P("I like to run" | N) = 0.000096$$

Logo, "I like to run" é classificada como positiva